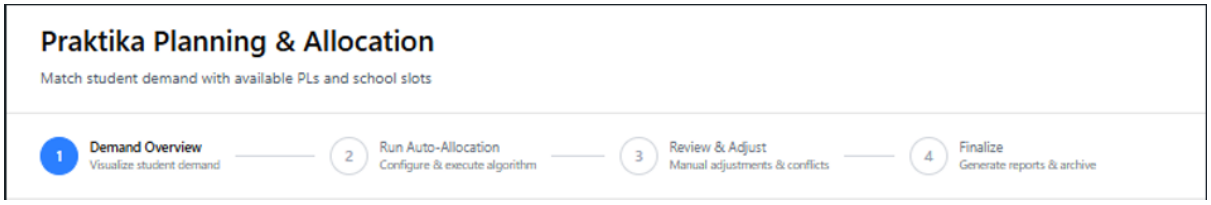


Analysis: Section 1 - The Stepper/Wizard Component

1. Verdict:
CORRECT. This stepper UI is the perfect high-level navigation for the allocation process. It correctly identifies the four key stages of the workflow: visualizing the problem, running the solver, reviewing the results, and finalizing.

2. What Needs Changing:

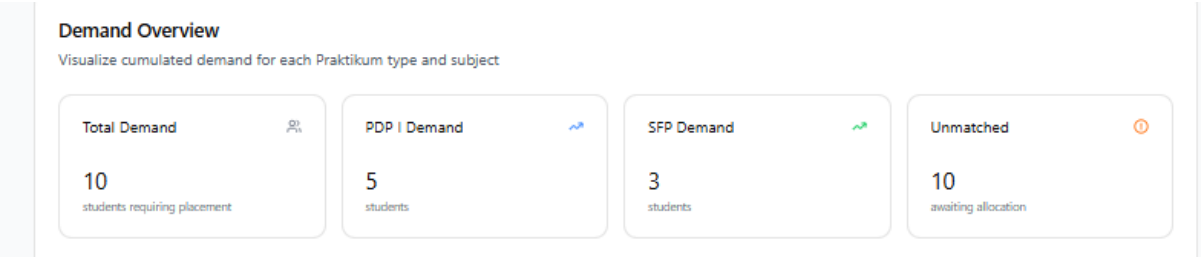
- **Nothing.** This component is conceptually perfect. The labels are clear and accurate.

3. Notes for Frontend Developer:

- This is a standard "Stepper" or "Wizard" component.
- The state for the activeStep (e.g., 1) should be managed in the main AllocationPage component.
- Clicking "Continue to Auto-Allocation" at the bottom of the page should simply update this state to 2.

4. API Response Required:

- **None.** This component is purely for UI state management and does not require its own API endpoint.



Analysis: Section 2 - KPI Summary Cards

1. **Verdict:**
   CONCEPTUALLY FLAWED. The flaw is in the "Total Demand" and "Unmatched" cards, as they do not provide actionable, independent information for the planning phase.
2. **What Needs Changing:**
   - **Remove Redundancy:** The "Total Demand" and "Unmatched" cards show the **exact same number** before the algorithm runs. The "Unmatched" card is confusing and should be removed.
   - **Add Supply/Capacity:** The primary goal of this screen is to assess readiness. The administrator *must* be able to see the total available mentor capacity.

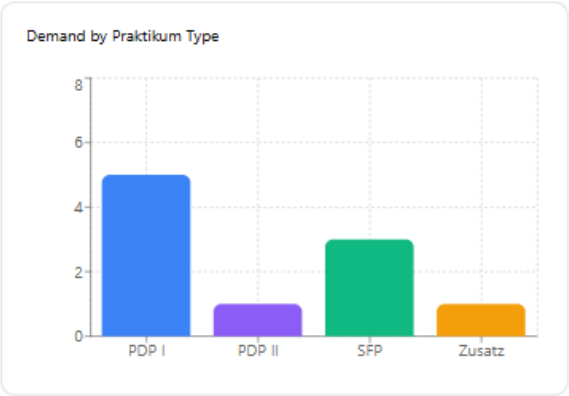
- **Complete the Picture:** The cards only show PDP I and SFP. They are missing PDP II and ZSP. All four must be represented.

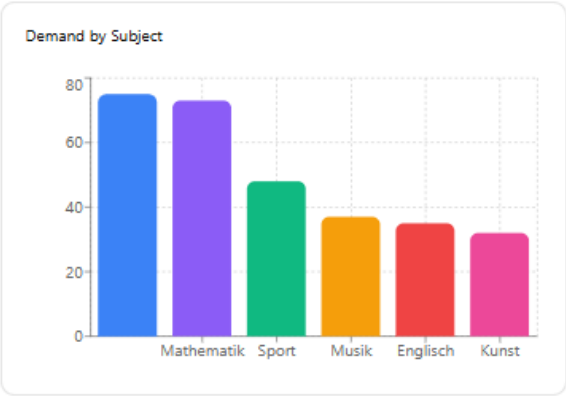
3. **How it should look like:**
   We will use the four cards to represent the most critical, high-level checks required before running the solver.

Card Position	Final Title	Value Calculation (API Source)	Subtext / Metric
Card 1	**Total Demand** (Problem Size)	SUM(required_slots) across all 4 types (PDP I/II, SFP/ZSP).	total_students requiring placement
Card 2	**PL Capacity**	Total PL Capacity (Anre-Std. * 2)	total mentor capacity slots
Card 3	**PDP Slots** (Block Focus)	required_slots for PDP I + PDP II.	X students total. (Focuses on the block internships for the year).
Card 4	**Wednesday Slots** (Flexibility Check)	required_slots for SFP + ZSP.	X slots total. (Focuses on the constrained Wednesday internships).

4. API response:

```
"summary_cards": {
  "total_demand_slots": 210,
  "total_pl_capacity_slots": 220,
  "total_pdp_demand": 95,     // Sum of required_slots for PDP I and PDP II
  "total_wednesday_demand": 115 // Sum of required_slots for SFP and ZSP
},
```





Analysis: Section 3 - The Bar Charts

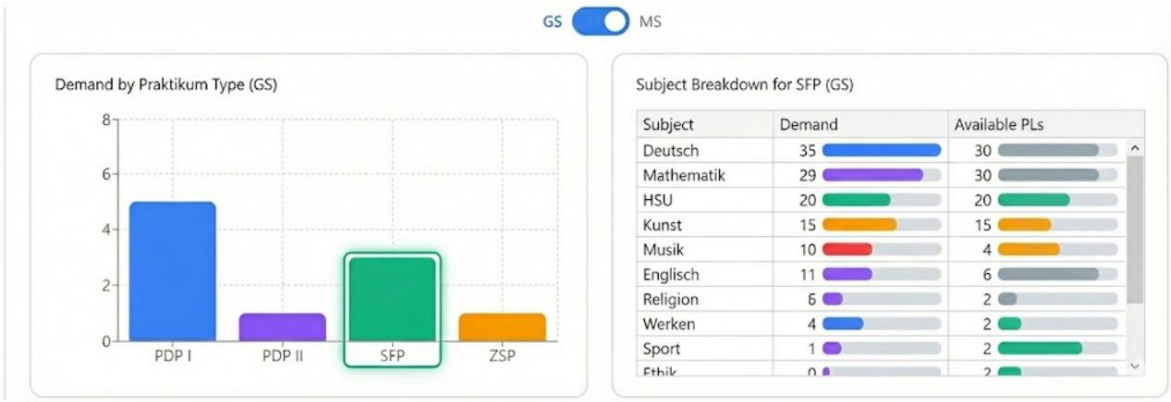
1. **Verdict:**
   CONCEPTUALLY FLAWED in the original prototype. While visualizing demand by type and subject is correct, the initial design is not scalable or specific enough for the administrator's needs.

2. **What Needs Changing:**

   - **Fix Terminology:** "Zusatz" must be renamed to **"ZSP"** for consistency.
   - **Separate GS/MS Data:** Demand for Grundschule (GS) and Mittelschule (MS) must be treated as two separate datasets, as they draw from different mentor pools.
   - **Improve Scalability:** A static "Top 5" subject chart is insufficient. The UI must handle a large number of subjects and allow the user to drill down for details.

3. Final Design (As shown in the new mockup):
The UI will consist of three linked components:

1. **A GS/MS Toggle Switch that controls the entire dataset.**
2. **An interactive "Demand by Praktikum Type" Bar Chart that acts as a filter.**
3. **A detailed "Subject Breakdown" Table that dynamically updates based on the selections in the toggle and the bar chart.**

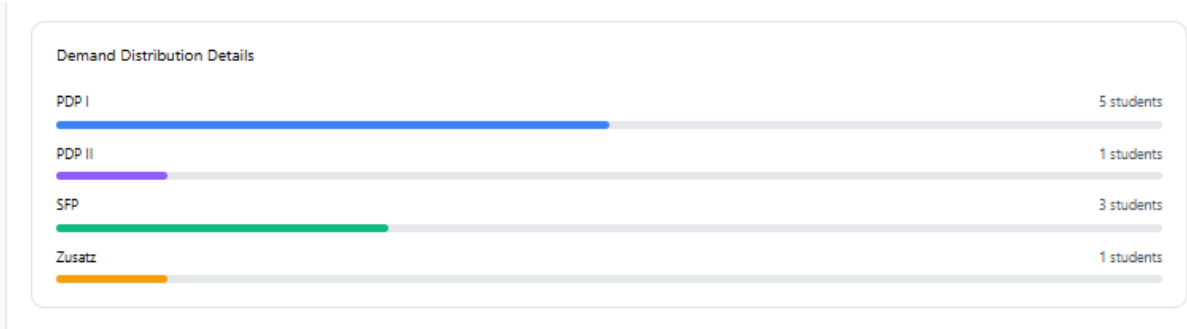


4. Notes for Frontend Developer:

- Implement a master `GS | MS` toggle. All data filtering starts here.
- The "Demand by Praktikum Type" chart's bars should be clickable. Clicking a bar (e.g., "SFP") updates a `selectedPraktikumType` state.
- The "Subject Breakdown" table's data is filtered based on *both* the `GS/MS` toggle and the `selectedPraktikumType`.
- The table's title should update dynamically (e.g., "Subject Breakdown for SFP (GS)").
- The table should compare `Demand` (number of students) directly against `Available PLs` (number of mentors).

5. API Response Required:

```
"detailed_breakdown": [
 {
  "practicum_type": "SFP",
  "program_type": "GS",
  "subject_code": "D",
  "required_slots": 35, // This is "Demand"
  "available_pls": 30   // This is "Available PLs"
 },
 {
  "practicum_type": "ZSP",
  "program_type": "GS",
  "subject_code": "HSU",
  "required_slots": 20,
  "available_pls": 20
 },
 // ... etc.
]
```



Analysis: Section 4 - The "Demand Distribution Details" Widget

1. **Verdict:**
   Its redundant of the demand by praktikum type char bar. It should be **deleted**.

Analysis: Section 5 - The Action Bar (Revised)

1. **Verdict:**
   SIMPLIFY. The "View Unmatched Demand" button, while potentially useful, is not essential for the core workflow, delete it.

Final API response:

GET /api/allocations/demand-preview


```
{
 "summary_cards": {
  "total_demand_slots": 210,
  "total_pl_capacity_slots": 220,
  "total_pdp_demand": 95,
  "total_wednesday_demand": 115
 },
 "detailed_breakdown": [
  {
/// examples
    "practicum_type": "PDP_I",
    "program_type": "GS",
    "subject_code": "N/A",
    "subject_display_name": "N/A",
    "required_slots": 50,
```

"available_pls": 55
},
{
"practicum_type": "PDP_II",
"program_type": "GS",
"subject_code": "N/A",
"subject_display_name": "N/A",
"required_slots": 45,
"available_pls": 48
},
{
"practicum_type": "SFP",
"program_type": "GS",
"subject_code": "D",
"subject_display_name": "Deutsch (D)",
"required_slots": 35,
"available_pls": 30
},
{
"practicum_type": "SFP",
"program_type": "MS",
"subject_code": "PCB",
"subject_display_name": "Physik/Chemie/Biologie (PCB)",
"required_slots": 8,
"available_pls": 10
},
{
"practicum_type": "ZSP",
"program_type": "GS",
"subject_code": "HSU",
"subject_display_name": "Heimat- und Sachunterricht (HSU)",

"required_slots": 20,

"available_pls": 20

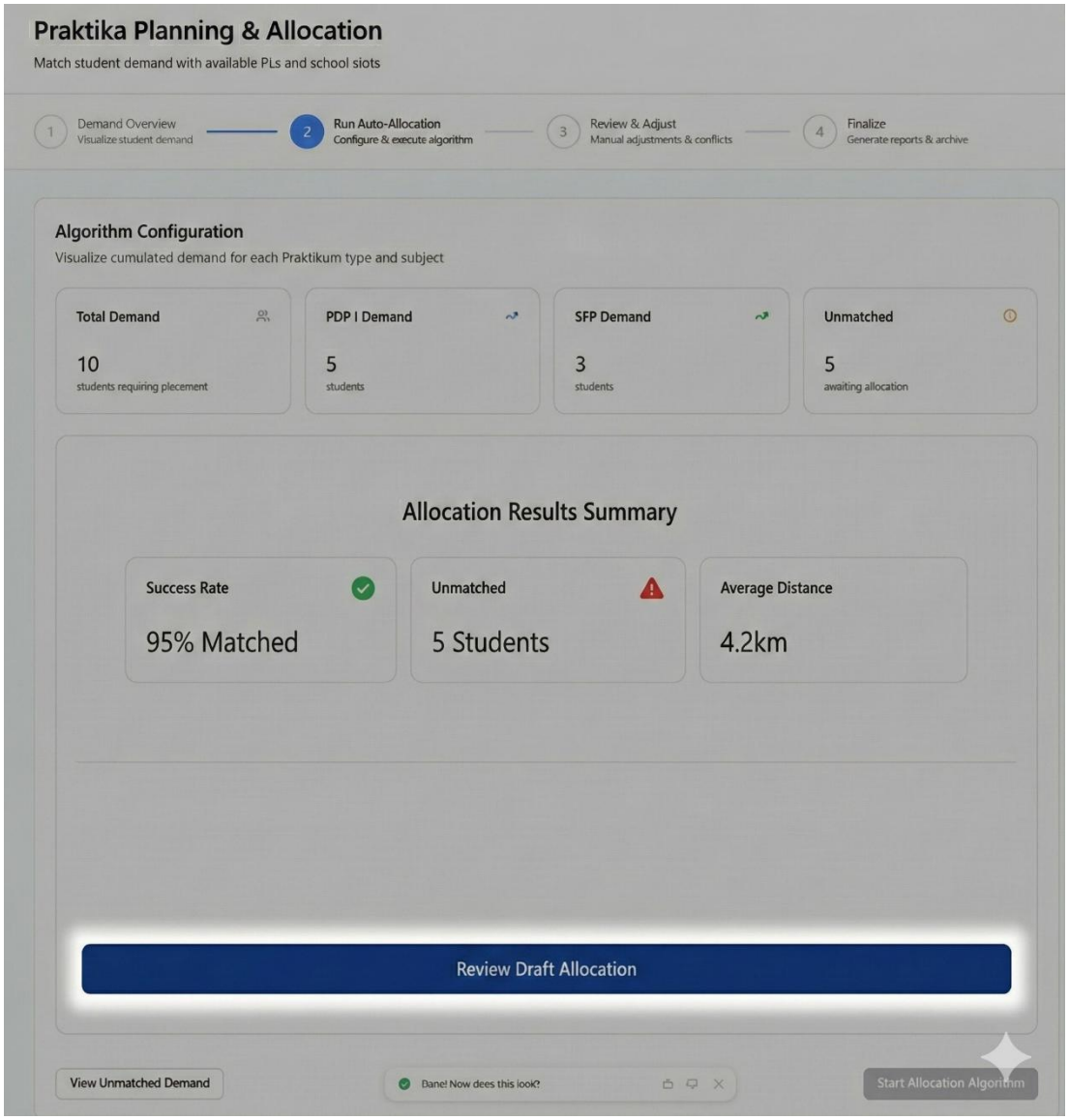
}

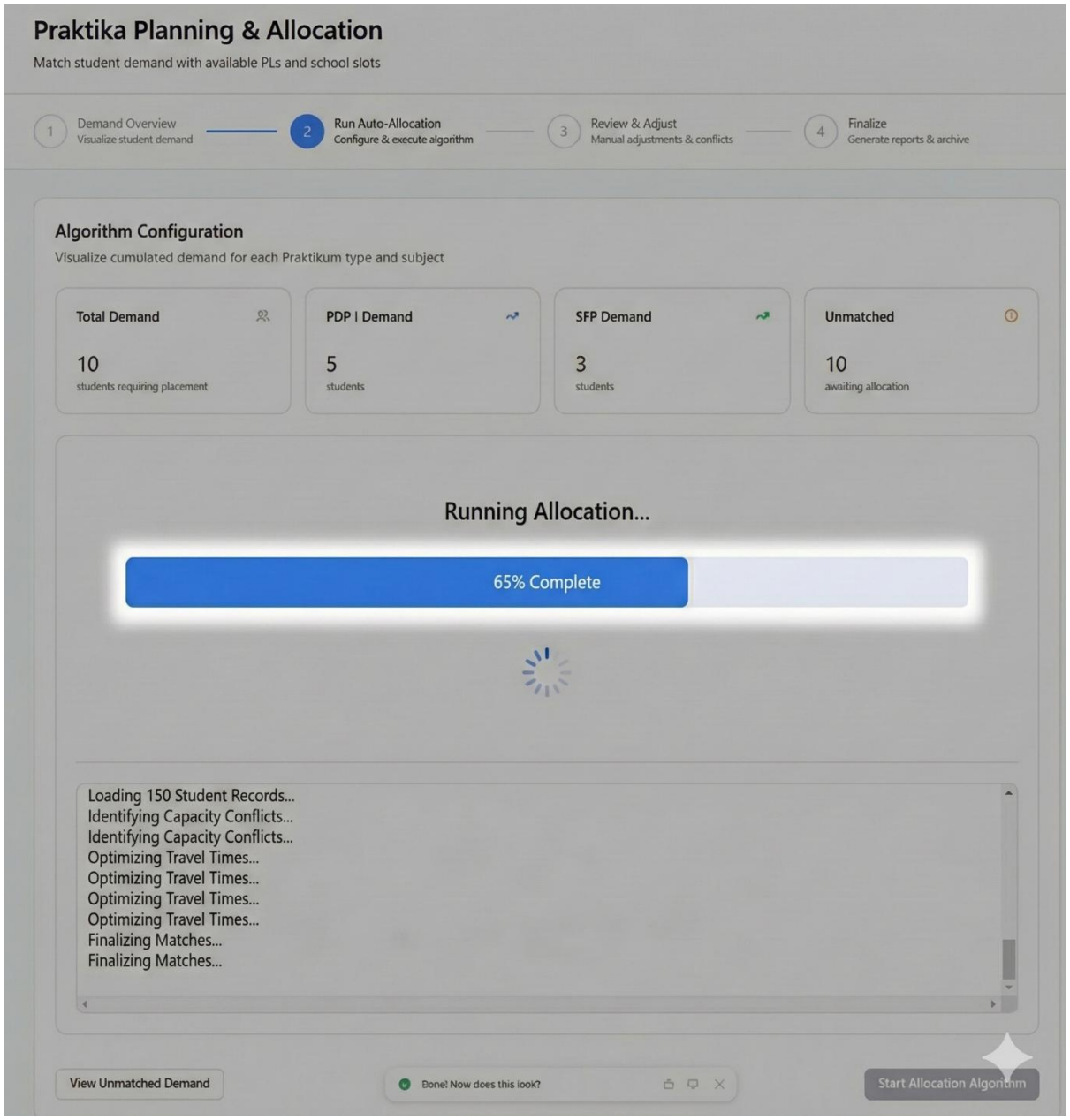
.....

]

}

Ideas for what happens when the *Start Allocation* button is pressed:


Praktika Planning & Allocation
Match student demand with available PLs and school siots
1 Demand Overview
Visualize student demand
2 Run Auto-Allocation
Configure & execute algorithm
3 Review & Adjust
Manual adjustments & conflicts
4 Finalize
Generate reports & archive
Algorithm Configuration
Visualize cumulated demand for each Praktikum type and subject
Total Demand
10
students requiring plecement
PDP I Demand
5
students
SFP Demand
3
students
Unmatched
5
awaiting allocation
Allocation Results Summary
Success Rate
95% Matched
Unmatched
5 Students
Average Distance
4.2km
Review Draft Allocation
View Unmatched Demand
Bane! Now dees this look?
Start Allocation Algorithm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Visualize cumulated demand for each Praktikum type and subject
Total Demand
10
students requiring placement
PDP I Demand
5
students
SFP Demand
3
students
Unmatched
10
awaiting allocation
Running Allocation...
65% Complete
Loading 150 Student Records...
Identifying Capacity Conflicts...
Identifying Capacity Conflicts...
Optimizing Travel Times...
Optimizing Travel Times...
Optimizing Travel Times...
Optimizing Travel Times...
Finalizing Matches...
Finalizing Matches...
View Unmatched Demand
Done! Now does this look?
Start Allocation Algorithm

Praktika Planning & Allocation

Match student demand with available PLs and school slots

1 Demand Overview — Visualize student demand
2 Run Auto-Allocation — Configure & executs algorithm
3 Review & Adjust — Manual adjustments & conflicts
4 Finalize — Generate reports & archive

Draft Allocation Review

Visualize cumulated demand for each Praktikum type and subject

Filter

Search

Student ID	Name	Praktikum Type	Subject	Allocated PL	Status	Actions
0000001	Anna Müller	PDP I	Deutsch	PL Weber	✓	Manual Adjust
0000002	Loura Hamert	PDP I	Deutsch	PL Weber	✓	Manual Adjust
0000003	Anma Mohnenn	SFP	Deutsch	PL Crotorrhuren	✓	Manual Adjust
0000003	Anna Müller	SFP	Deutsch	PL Weber	⚠ Potential Conflict	Manual Adjust
0000004	Maria Hamerz	PDP I	Deutsch	PL Weber	✓	Manual Adjust
0000005	Stetan Verkerz	PDP I	Fìm	PL Weber	✓	Manual Adjust
0000006	Anna Müller	SFP	Deutsch	PL Weber	✓	Manual Adjust
0000007	Jana Müller	SFP	Deutsch	PL Weber	✓	Manual Adjust
0000008	Andrea Somaren	PDP I	Deutsch	PL Weber	✓	Manual Adjust
0000010	Anna Müller	PDP I	Deutsch	PL Weber	✓	Manual Adjust

1 2 3

Dane? Now dees this iook?

Finalize & Generate Reports